On my honor, I have neither given nor received any unauthorized and/or inappropriate assistance for all sessions of this exam. The work done on this exam is totally my own. I understand that by the school code, violation of these principles will lead to a zero grade and is subject to harsh discipline issues.

B.ilbey Gontar /15072004Y

(BG)

① $\begin{bmatrix}1&0&0&0\\-2&1&0&0\\m&0&1&0\\-1&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0\\0&1&0&-8\\m&0&1&0\\-1&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0\\0&1&0&-8\\m&0&1&0\\0&0&0&1\end{bmatrix}\begin{bmatrix}1&0&0&0\\0&1&0&0\\m-2&0&1&2\\0&0&0&0\end{bmatrix}$ $\begin{matrix}m=2\\k=-\frac{1}{2}\\\alpha=2\end{matrix}$ $\begin{matrix}\beta=0\\\gamma=1/3\end{matrix}$

② a- Minimum dimension of space = 2 because it has to be 2Δ for make a vector.

b-

c-

③ $H=I-2u.u^T$ $\quad u.u^T=I \quad H=-I$

a- $I=A.A^T \rightarrow H^2=(-I)^2=I$

b- H is orthonormal because ⊙ square matrix ⊙ $u.u^T=I$ ⊙ $A^{-1}=A^T$

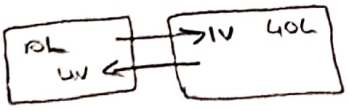
c-

d- Yes H is diagonalizable cause of (square structure) symmetric matrix.

④ a- $\begin{bmatrix}v_{k+1}\\w_{k+1}\end{bmatrix}=\begin{bmatrix}-2&1\\1&-2\end{bmatrix}\begin{bmatrix}v_k\\w_k\end{bmatrix}$ → differential equation set

$\det A = 1/4$

b-



$\frac{4V}{13V}=\frac{x}{100} \cong \%30$

c-

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